# 9.2 Resistance

## Question Paper

| | |
|---|---|
| Course | CIE A Level Physics |
| Section | 9. Electricity |
| Topic | 9.2 Resistance |
| Difficulty | Medium |

**Time allowed:** 40

**Score:** /27

**Percentage:** /100

## Question 1a

The photograph shows a statue of Buddha in Sri Lanka, which is protected by a lightning conductor.

© Valery Shanin/123RF

During a storm, a potential difference of 2.7 MV was generated between a cloud and the top of the lightning conductor on the statue. A flash of lightning passed between the cloud and the lightning conductor, producing a current of 25 kA for a time of 7.5 ms.

Calculate the energy transferred by the lightning strike.

Energy transferred = ..........................................

**[3 marks]**

## Question 1b

The lightning conductor is made from copper.

Sketch an *I–V* graph for the rod when it conducts a lightning strike to earth

**[3 marks]**

## Question 1c

Explain any changes which the lightning rod would undergo during a lightning strike.

**[3 marks]**

## Question 2a

Fig. 1.1. shows the graph of V against I for a non-ohmic conductor.

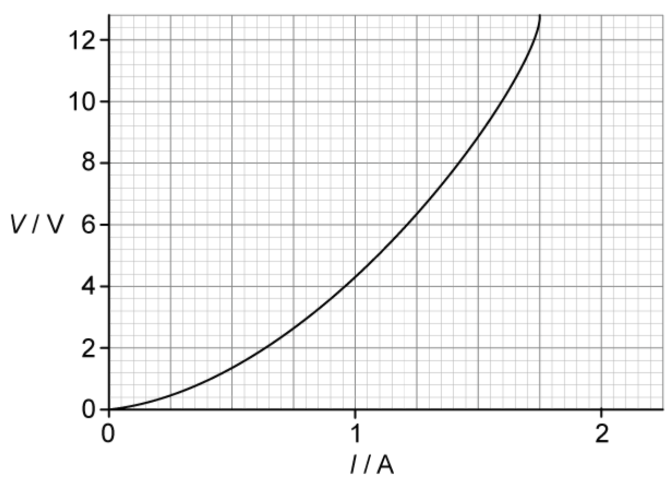



**Fig. 1.1.**

Calculate the maximum resistance of the lamp over the range shown by the graph.

**[3 marks]**

## Question 2b

Using the axes in Fig. 1.2. sketch the graph of current against potential difference for a diode.

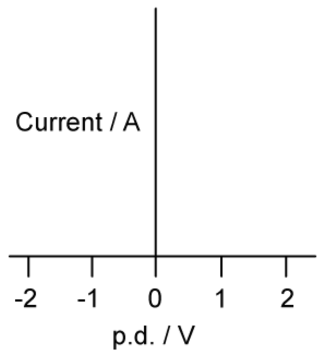



**Fig. 1.2.**

**[2 marks]**

## Question 2c

For the graph in Fig. 1.3. determine

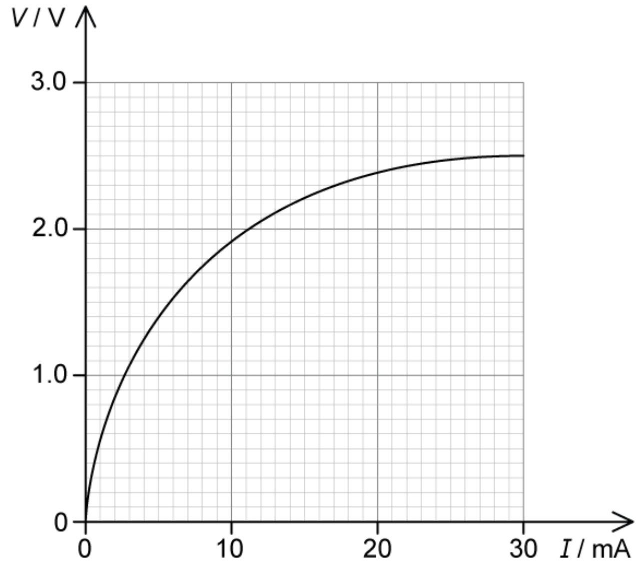



**Fig. 1.3.**

i)

the resistance when current = 10 mA

[2]

ii)

the power when current = 10 mA

[1]

**[3 marks]**

## Question 3a

An electrical component has the I-V characteristic shown in Fig. 1.1.

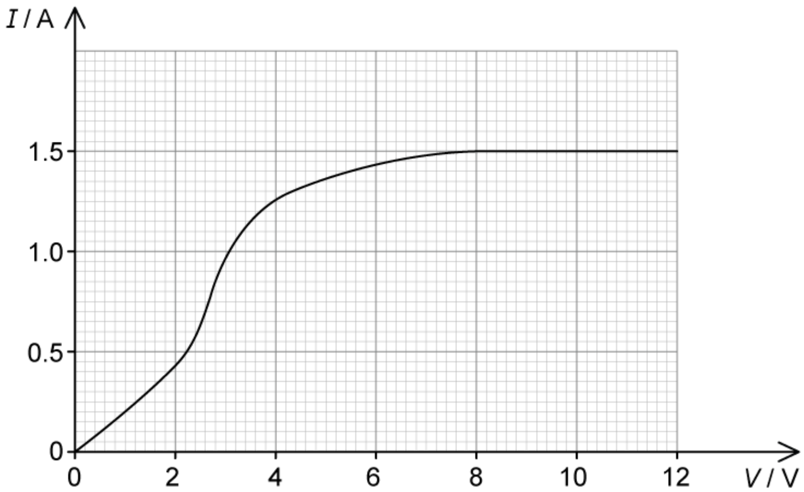



**Fig. 1.1**

Determine the resistance of the component when a potential difference of 6.0 V is applied across it.

**[2 marks]**

## Question 3b

For the component in part **(a)** calculate the minimum value of *R* as the potential difference across it changes over the range 0–10 V.

**[2 marks]**

## Question 3c

For the component in part **(a)** calculate

(i)
The power dissipated when $V = 2.0$ V

[2]

(ii)
The maximum power dissipated before the component heats up

[2]

(iii)
The value of resistance at the point identified in part (ii)

[2]

**[6 marks]**

# Model Answers: Medium

1a

a) Calculate the energy transferred by the lightning strike:

List the known quantities:

- Potential difference, $V$ = 2.7 MV = 2.7 × $10^6$ V
- Current, $I$ = 25 kA = 25 × $10^3$ A
- Time, $t$ = 7.5 ms = 7.5 × $10^{-3}$ s

Use the equation defining current to calculate the charge, $Q$:

- $Q = I t = (25 \times 10^3) \times (7.5 \times 10^{-3})$
- $Q = 187.5$ C [1 mark]

Use the equation defining potential difference to calculate work done, $W$:

- $V = \frac{W}{Q}$
- $W = (2.7 \times 10^6) \times 187.5$ [1 mark]
- $W = 5.0625 \times 10^8 = 5.1 \times 10^8$ J (2 s.f.) [1 mark]

**[Total: 3 marks]**

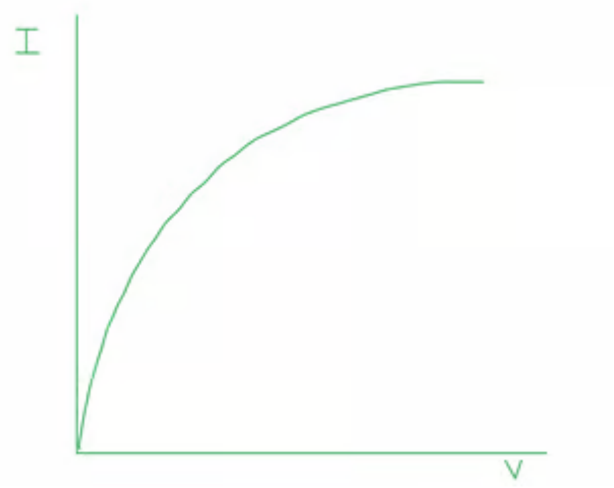
1b

b) The lightning conductor is made from copper. Sketch an IV graph for the rod when it conducts a lightning strike to earth:

Decide which type of conductor the rod is:

- (The rod will get hot since copper wires get hot, therefore) it is the equivalent of a non-ohmic resistor / filament lamp; [1 mark]

A sketch of the correct graph shape looks like:




- Axes labelled *I* and *V* (either way round); [1 mark]
- Correct graph shape for chosen axes; [1 mark]

**[Total: 3 marks]**

If you attempted to draw **any** version of the filament lamp graph, even if it went a bit wrong, the examiner would award you the first mark.

1c

c) Explain any changes which the lightning rod would undergo during a lightning strike:

- Delocalised / free electrons move faster / increase kinetic energy; [1 mark]
- Therefore, as current increases the temperature increases; [1 mark]
- As temperature increases the resistance increases (as shown in the graph); [1 mark]

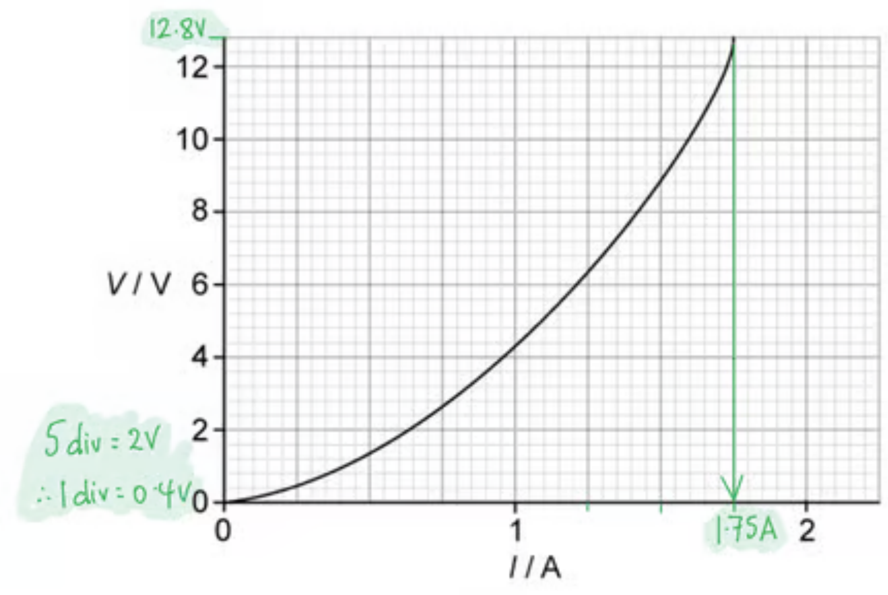
**[Total: 3 marks].**

2a

a) Calculate the maximum resistance of the lamp over the range shown by the graph:

Identify the point of maximum resistance on the graph and choose *V* and *I* accordingly:

- resistance = gradient of a *VI* graph, therefore find steepest part of slope




- V = 12.8 V **AND** I = 1.75 A [1 mark]

Use the equation for Ohm's Law:

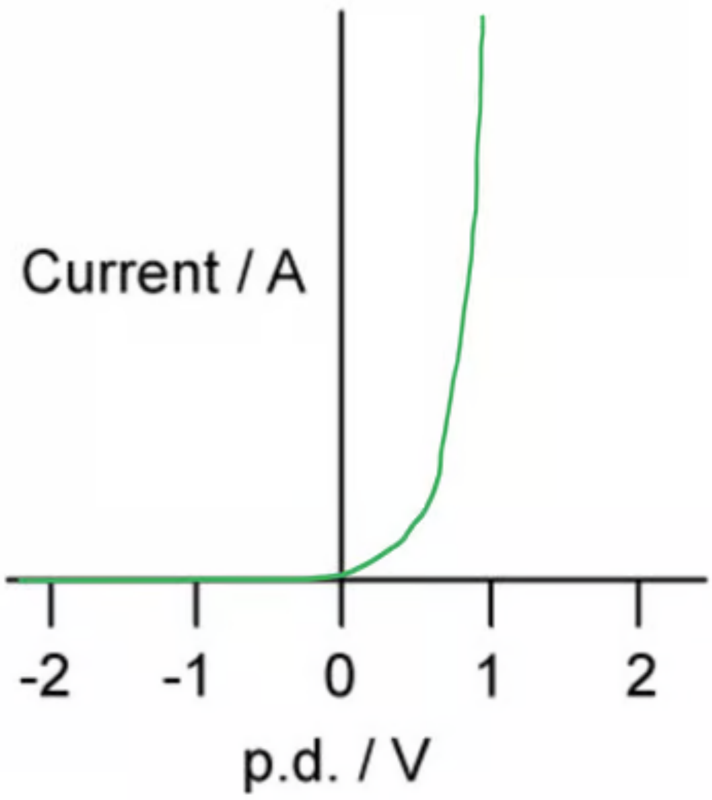
- $V = I R$
- $R = \frac{V}{I} = \frac{12.8}{1.75}$ [1 mark]
- $R = 7.3\ \Omega$ [1 mark]

**[Total: 3 marks]**

2b

b) Using the axes in Fig. 1.2. sketch the graph of current against potential difference for a diode:

A graph which scores **two marks** looks like:




- Current zero for all negative voltages; [1 mark]
- For positive voltages line slopes up gently to about 0.6 V **AND** then has very steep rise between 0.6 - 1.0 V; [1 mark]

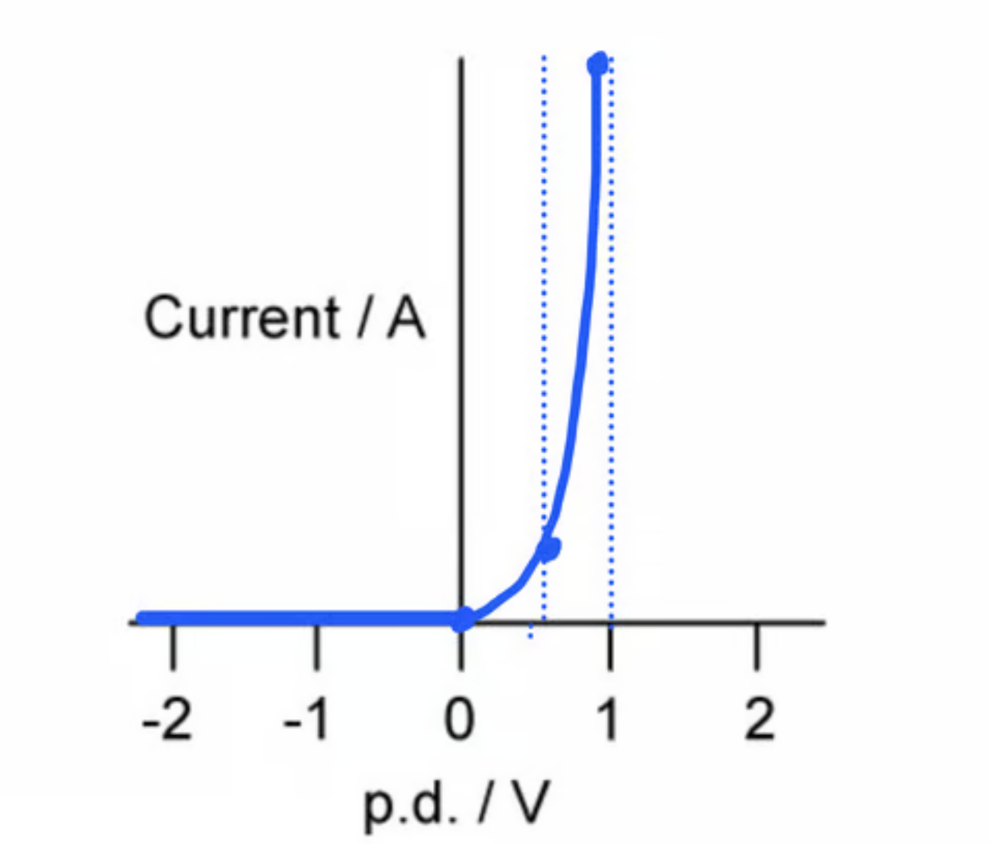
**[Total: 2 marks]**

It's a good idea to sketch out graph shapes in pencil first, so your final answer is correct. Our sketch above started off looking like this, with guidelines, points to pass through **and two** drafts of the graph, before we rubbed out all the preparation and mistakes!

- Current zero for all negative voltages; [1 mark]
- For positive voltages line slopes up gently to about 0.6 V **AND** then has very steep rise between 0.6 - 1.0 V; [1 mark]

**[Total: 2 marks]**

It's a good idea to sketch out graph shapes in pencil first, so your final answer is correct. Our sketch above started off looking like this, with guidelines, points to pass through **and two** drafts of the graph, before we rubbed out all the preparation and mistakes!




2c

(c)

i) The resistance when current = 10 mA:

Determine the corresponding value of potential difference from the graph:

- When $I$ = 10 mA, V = 1.9 V

Calculate the resistance:

- $R = \frac{V}{I} = \frac{1.9}{10 \times 10^{-3}}$ [1 mark]

- $R = 190\ \Omega$ [1 mark]

ii) The power when current = 10 mA:

Use the power equation which relates current and resistance:

- $P = I^2 R = (10 \times 10^{-3})^2 \times 190 = 1.9 \times 10^{-2}$ W **or** 19 mW [1 mark]

**[Total: 3 marks]**

3a

(a) Determine the resistance of the component when a potential difference of 6.0 V is applied across it:

Use the graph to find current when p.d. = 6.0 V:

- Corresponding value of *I* = 1.425 A

Use Ohm's law equation to find resistance:

- $V = IR \Rightarrow R = \frac{V}{I}$

- $R = \frac{6.0}{1.425}$ [1 mark]

- $R = 4.2\ \Omega$ [1 mark]

**[Total: 2 marks]**

The graph is linear at 6.0 V, so the corresponding value can be read off.

It is encouraged, for more difficult scales, to draw a line using a pencil and ruler to the *y* axis to read off the most accurate value.

3b

(b) For the component in part (a) calculate the minimum value of $R$ as the potential difference across it changes over the range 0–10 V:

Determine where the answer lies on the graph:

- Least value of $R$ is found where the gradient is steepest

Read the values of $V$ and $I$ at the point where the gradient is steepest and calculate:

- $R = \frac{V}{I} = \frac{3.0}{0.95}$ [1 mark]

- $R = 3.0\ \Omega$ [1 mark]

**[Total: 2 marks]**

Graphs of current and potential difference are commonly plotted as $V$-$I$ and $I$-$V$, so you need to check rather than just memorising the shapes.

Since the resistance is $\frac{V}{I}$ and the gradient is $\frac{I}{V}$, this means the resistance is $\frac{1}{gradient}$.

3c

(c) For the component in part (a) calculate:

(i) The power dissipated when $V$ = 2.0 V:

Determine the current when $V$ = 2.0 V by finding the corresponding value on the graph:

- Current, $I$ = 0.425 A [1 mark]

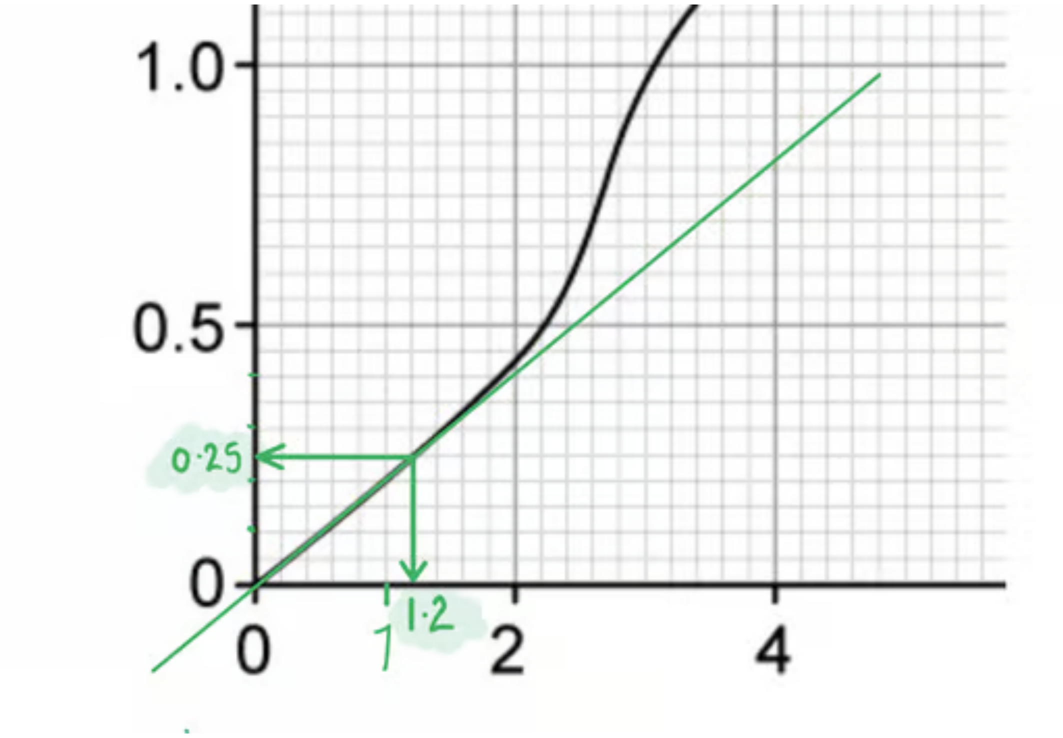
Use the equation for power:

- $P = VI = 2.0 \times 0.425 = 0.85$ W [1 mark]

(ii) The maximum power dissipated before the component heats up:

Use the graph to find where the component starts to heat up:

- Until the temperature changes, the component obeys Ohm's Law
- Therefore, identify the last point of the straight line / directly proportional area **OR** clear use of correct values / point marked on the graph; [1 mark]




Determine the current and potential difference at this point:

- Current, *I* = 0.25 A
- Potential difference, *V* = 1.2 V

Calculate using the equation for power:

- $P = VI = 1.2 \times 0.25 = 0.30 \text{ W}$ [1 mark]

(iii) The value of resistance at the point identified in part (ii):

Use an equation relating power and resistance to find the value of *R*:

- $P = I^2 R$ **or** $P = \frac{V^2}{R}$
- $R = \frac{P}{I^2} = \frac{0.30}{0.25^2}$ **or** $R = \frac{V^2}{P} = \frac{1.2^2}{0.30}$ [1 mark]

- $R = 4.8\ \Omega$ [1 mark]

**[Total: 6 marks]**

A question like this, which depends on correctly reading a set of values from a graph, would always have the subsequent marks as 'error carried forward'.

So if you read the graph incorrectly, but then continued with the calculation as you should have, then you'll have a different answer to ours, but still potentially get the marks.